

Laborator 5

Rutare Dinamica (OSPF)

Scopul acestui laborator:

- Configurarea protocolului OSPF
- Intelegerea rutarii automate
- Observarea vecinilor, a redundantei si a convergentei
- Intelegerea diferentei fata de rutare statica

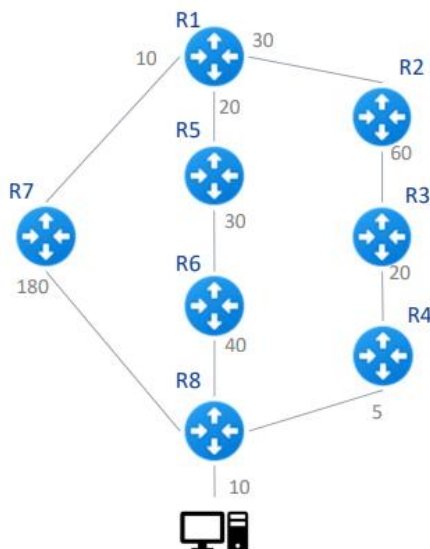
Idee de baza:

- Fiecare LAN = o retea separata
- Routerule le conecteaza
- Daca laboratorul trecut am scris manual rutele prin *ip route*
- Acum routerule isi spun singure retelele
 - Isi construiesc automat tablea de rutare
- Fiecare router avand 2 vecini, exista doua drumuri posibile intre retele
 - Daca un link cade -> OSPF gaseste alt drum

Scurta recapitulare RC:

Rutare dinamică

Rip vs OSPF



RIP – Routing Information Protocol

RIP – ia în calcul doar numărul de hop-uri de la sursă la destinație

OSPF – Open Shortest Path First

OSPF – ia în calcul un “cost” al legăturii ce ține cont de mai mulți parametri (trecute cu gri în figură)

Ruta	Localizare	Cost cumulat OSPF
R1-R7-R8	Stanga	$10+180+10=200$
R1-R5-R6-R8	Mijloc	$20+30+40+10=100$
R1-R2-R3-R4-R8	Dreapta	$30+60+20+5+10=125$

Astfel:

- Dacă am folosi protocolul RIP am parcurge calea din stânga
- Dacă folosim OSPF folosim calea din mijloc

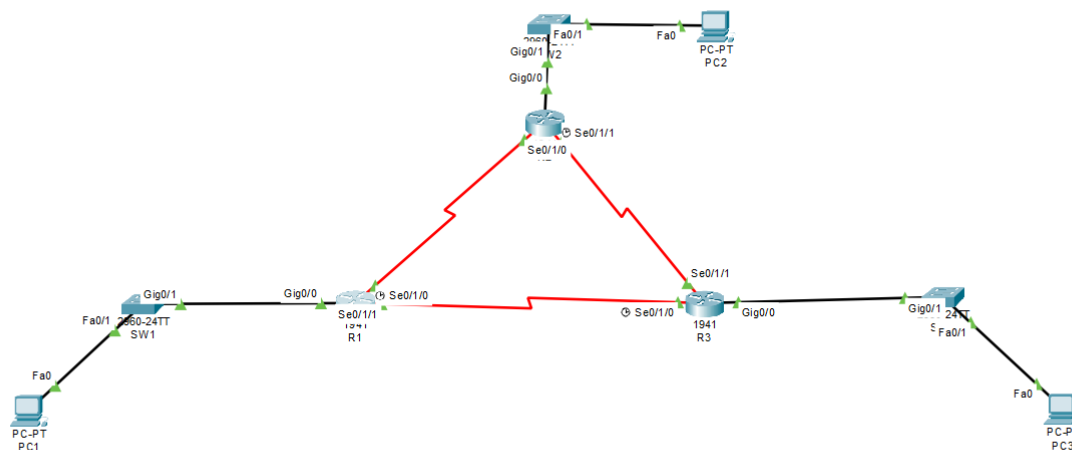
1. Topologie si schema logica:

Dispozitive:

- 3 Switch-uri
- 3 Routere
- 3 PC-uri

Conexiuni:

De la	La	Porturi	Cablu (tip)
PC1	SW1	Fa0 -> Fa0/1	Straight
SW1	R1	Gi0/1 -> Gi0/0	Straight
PC2	SW2	Fa0 -> Fa0/1	Straight
SW2	R2	Gi0/1 -> Gi0/0	Straight
PC3	SW3	Fa0 -> Fa0/1	Straight
SW3	R3	Gi0/1 -> Gi0/0	Straight
R1	R2	Se0/1/0 ↔ Se0/1/0	Serial DCE
R2	R3	Se0/1/1 ↔ Se0/1/1	Serial DCE
R1	R3	Se0/1/1 ↔ Se0/1/0	Serial DCE



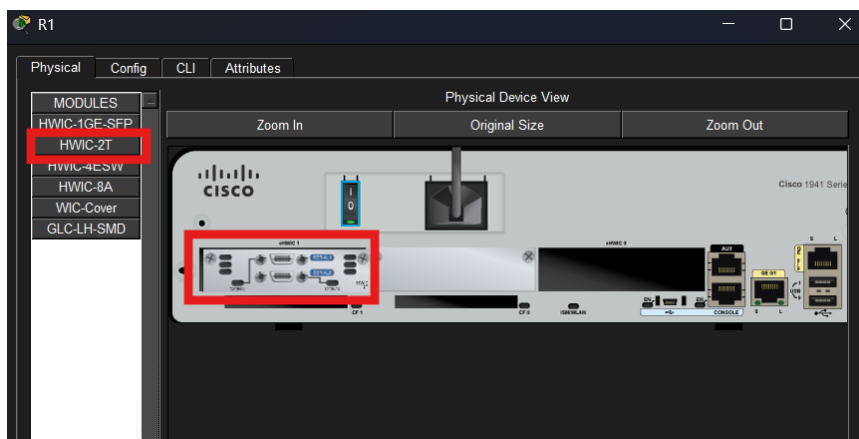
PORTUL SERIAL:

Vom adauga modulul **HWIC-2T**.

Este un modul care ofera 2 porturi serial, se va monta in router (ca un „card” hardware) si permite conexiuni **Serial DCE/DTE** intre routere.

DCE- Data Circuit-terminating Equipment, DTE- Data Terminal Equipment.

1. Dam click pe router, mergem la tab-ul Physical.
2. Oprim routerul, click pe butonul de Power Off.
3. Adaugam modulul, din lista de module din stanga, tragem HWIC-2T in slotul liber dedicat.
4. Pornim routerul.



Folosim cablul Serial DCE. Un router o sa fie DCE si celalalt devine automat DTE.

! Pe routerul DCE, trebuie sa setam clock rate. (veti vedea dupa cum)

2. Plan de adresare

Dispozitiv	IP Address	Default Gateway
PC1	192.168.10.2/24	192.168.10.1
PC2	192.168.20.2/24	192.168.20.1
PC3	192.168.30.2/24	192.168.30.1

Legatura dintre routere:

Vom folosi 10.0.0.x deoarece este o retea separata doar pentru routere.

Legaturi	IP Address	IP Address
R1-R2	10.0.0.1/30	10.0.0.2/30
R2-R3	10.0.0.5/30	10.0.0.6/30
R1-R3	10.0.0.9/30	10.0.0.10/30

Explicatie:

Intre doua routere conectate direct avem nevoie de:

- retea
- 2 adrese IP
- una pentru primul router, una pentru al doilea

Pentru legaturile dintre routere folosim retele foarte mici, de tip /30. La /30, retelele cresc din 4 in 4.

Pentru fiecare legatura seriala vom face asa:

- alegi o retea
- dai primul IP unui router
- dai al doilea IP celui alt router

Noi avem 3 legaturi intre routere:

R1 ↔ R2

R2 ↔ R3

R1 ↔ R3

Fiecare legatura are reteaua ei proprie.

Link 1: R1 - R2

Reteaua folosita: 10.0.0.0/30

IP-uri folosite: -R1: 10.0.0.1

-R2: 10.0.0.2

De retinut:

10.0.0.0 = numele retelei

10.0.0.1 si 10.0.0.2 = IP-uri de pe routere

Link 2: R2 - R3

Reteaua folosita: 10.0.0.4/30

IP-uri folosite: -R2: 10.0.0.5

-R3: 10.0.0.6

Link 3: R1 - R3

Reteaua folosita: 10.0.0.8/30

IP-uri folosite: - R1: 10.0.0.9

-R3: 10.0.0.10

3. Configurare PC-uri

Pe fiecare PC:

Desktop → IP Configuration

Adaugati IP Address, Subnet Mask si Default Gateway.

4. Configurarea Switch1

Port catre PC1:

```
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#no shutdown
```

Port catre R1:

```
Switch(config-if)#interface gig0/1
Switch(config-if)#switchport mode access
Switch(config-if)#no shutdown
```

Salvare:

copy running-config startup-config

5. Configurarea Switch2

La fel ca la SW1.

6. Configurarea R1

LAN catre PC1:

```
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
```

Interfata seriala catre R2:

```
Router(config)#interface serial 0/1/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
```

Clock rate- seteaza viteza fizica de transmitere a datelor (bps).

Interfata seriala catre R3:

```
Router(config-if)#interface serial 0/1/1
Router(config-if)#ip address 10.0.0.9 255.255.255.252
Router(config-if)#no shutdown
```

Configurare OSPF:

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#network 10.0.0.8 0.0.0.3 area 0
Router(config-router)#exit
```

Explicatie:

Porneste OSPF 1, anunta LAN1 in OSPF, anunta *Linkul R1-R2* in OSPF si anunta *Linkul R1-R3* in OSPF.

Area 0 – zona de backbone si toate routerele trebuie sa fie in aceeasi arie in acest laborator.

0.0.0.3 – **wildcard mask**

OSPF nu foloseste subnet mask (255.255.255.252), ci foloseste inversul mastii (wildcard mask). Adica ia in calcul doar 4 adrese (daca retea e 10.0.0.0, va lua in calcul de la 10.0.0.0 la 10.0.0.3)

7. Configurarea R2

LAN catre PC2:

```
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
```

Interfata seriala catre R1:

```
Router(config-if)#interface serial 0/1/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
```

Interfata seriala catre R3:

```
Router(config-if)#interface serial 0/1/1
Router(config-if)#ip address 10.0.0.5 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
```

Configurarea OSPF:

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#
05:45:46: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.1 on Serial0/1/0 from LOADING to FULL, Load
Done
Router(config-router)#network 10.0.0.4 0.0.0.3 area 0
Router(config-router)#exit
```

8. Configurarea R3

LAN catre PC3:

```
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown
```

Interfata seriala catre R2:

```
Router(config-if)#interface serial 0/1/1
Router(config-if)#ip address 10.0.0.6 255.255.255.252
Router(config-if)#no shutdown
```

Interfata seriala catre R1:

```
Router(config-if)#interface serial 0/1/0
Router(config-if)#ip address 10.0.0.10 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
```

Configurarea OSPF:

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 10.0.0.4 0.0.0.3 area 0
Router(config-router)#netw
05:53:59: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.20.1 on Serial0,
Done

% Incomplete command.
Router(config-router)#network 10.0.0.8 0.0.0.3 area 0
-
```

Salvare pe routere:

copy running-config startup-config

9. Verificarea configuratiei

Pe fiecare router:

- Verificarea interferelor:

show ip interface brief

- Verificarea vecinilor OSPF:

show ip ospf neighbor

- Tabela de rutare:

show ip route

10. Demonstratie convergenta

Opriti un link serial si observati daca reteaua continua sa functioneze.

Ex: legatura directa dintre R1-R3:

```
Router(config)#interface serial 0/1/1
Router(config-if)#shutdown
```

```
Router(config-if)#
```

```
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to administratively down
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down
```

```
06:04:54: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on Serial0/1/1 from FULL to DOWN, Neighbor Down: Interface down or detached
```

```
Router(config-if)#end
```

Testati.

Ping-ul ar trebui sa continue sa functioneze dupa convergenta, deoarece traficul va merge pe ruta alternativa: R1 → R2 → R3.